LATIHAN 3

Tentukan $\frac{dy}{dx}$ jika diketahui

1. $y = 2 + \frac{1}{x^2}$
2. $y = (x^3 + 7x^2 - 8)\left(\frac{\sqrt{10}}{x^7}\right)$
3. $y = \left(\frac{1}{x} + \frac{1}{x^2}\right)(3x^3 + 27)$
4. $y = (x^5 + 2x)^2$
5. $y = \frac{x^5}{x^3 - 2}$
6. $y = (2x^7 - x^2)\left(\frac{x-1}{x+1}\right)$
7. $y = x^{-5}(2x^9 + 1)$
8. $y = (x^2 + 2x)(4 - 3x)$
9. $y = \frac{\sec x}{1 + \tan x}$
10. $y = \frac{\csc x}{\tan x}$
11. $y = \frac{\sin x \sec x}{1 + x \tan x}$
12. $y = \frac{(x^2+1)\cot x}{3 - \cos x \csc x}$

13. $y = (x^3 + 4x)^{15}$

14. $y = (3x^2 + 2x - 1)^6$

15. $y = \left(x^3 - \frac{7}{x}\right)^{-2}$

16. $y = \frac{1}{(x^4+x+1)^9}$

17. $y = \frac{4}{(3x^2-2x+1)^3}$

18. $y = \sqrt{x^3 - 2x + 5}$

19. $y = \sqrt{4 + 3\sqrt{x}}$

20. $y = \sin^3 x$

21. $y = \sin\left(\frac{1}{x^2}\right)$

22. $y = \tan^4(x^3)$

23. $y = 2\sec^2(x^7)$

24. $y = \cos^3\left(\frac{x}{x+1}\right)$

25. $xy + 2x + 3x^2 = 4$

26. $\frac{1}{x} + \frac{1}{y} = 1$

27. $\sqrt{x} + \sqrt{y} = 4$

28. $\frac{y}{x-y} = x^2 + 1$

29. $x^2 - y^2 = 100$

30. $x^3 - y^3 = 6xy$

31. $x^2y + 3xy = 3x$

32. $\frac{xy^3}{1+\sec y} = 1 + y^4$

33. $\sin x + \cos y = \sin x \cos y$

34. $x^2 = \frac{\cot y}{1+\csc y}$

35. $\cos xy = y$

36. $\sin(x^2y^2) = x$